K-Fold Cross-Validation: Problems with Detailed Solutions

Problem 1: Basic K-Fold Cross-Validation (5 Marks)

Question: A machine learning dataset contains 100 student records. A researcher evaluates a classification model using 5-fold cross-validation.

1. How many folds will be created?
2. How many records will be used for training in each iteration?
3. How many records will be used for testing in each iteration?
4. How many times will the model be trained?
5. Calculate the average accuracy if the fold accuracies are 85%, 90%, 88%, 92%, and 85%.

Solution: Given: Total records = 100 and number of folds (K) = 5.

1. Number of folds: The dataset is divided into 5 equal folds.

2. Testing records per fold: 100 / 5 = 20 records.

3. Training records per fold: 100 – 20 = 80 records.

4. Number of training iterations: The model is trained 5 times, with a different fold used for testing in each iteration.

5. Average accuracy:
Average Accuracy = (85 + 90 + 88 + 92 + 85) / 5 = 440 / 5 = **88%**.

Problem 2: Calculate Average Accuracy (5 Marks)

Question: A dataset contains 120 records and is evaluated using 4-fold cross-validation. The accuracies obtained are 80%, 85%, 90%, and 85%. Calculate the average accuracy, highest and lowest accuracy, difference, and training/testing records per fold.

Solution: Given: Total records = 120 and K = 4.

1. Testing records per fold: 120 / 4 = 30 records.

2. Training records per fold: 120 – 30 = 90 records.

3. Average accuracy: (80 + 85 + 90 + 85) / 4 = 340 / 4 = **85%**.

4. Highest and lowest accuracy: Highest = 90% (Fold 3); lowest = 80% (Fold 1).

5. Difference: 90 – 80 = **10 percentage points**.

Final answers: Training = 90 records; testing = 30 records per iteration; average accuracy = 85%; highest = 90%; lowest = 80%; difference = 10 percentage points.

Problem 3: Compare 5-Fold and 10-Fold Cross-Validation (5 Marks)

Question: A dataset contains 200 records. Compare 5-fold and 10-fold cross-validation.

Solution:

Case 1: 5-Fold Cross-Validation

Testing records = 200 / 5 = 40.
Training records = 200 – 40 = 160.
The model is trained 5 times.

Case 2: 10-Fold Cross-Validation

Testing records = 200 / 10 = 20.
Training records = 200 – 20 = 180.
The model is trained 10 times.

Parameter	5-Fold CV	10-Fold CV
Number of folds	5	10
Testing records per fold	40	20
Training records per fold	160	180
Model training runs	5	10

Conclusion: 10-fold cross-validation uses more records for training in each iteration, but it requires twice as many training runs as 5-fold cross-validation, increasing computational cost.

Problem 4: K-Fold Cross-Validation with Error Rates (5 Marks)

Question: A classification model is evaluated using 5-fold cross-validation on 500 records. The error rates are 8%, 10%, 6%, 12%, and 9%. Calculate the average error rate, average accuracy, and best and worst folds.

Solution:

1. Average error rate: (8 + 10 + 6 + 12 + 9) / 5 = 45 / 5 = **9%**.

2. Average accuracy: 100% – 9% = **91%**, assuming accuracy and error rate are complementary.

3. Best-performing fold: Fold 3, with the lowest error rate of 6%.

4. Worst-performing fold: Fold 4, with the highest error rate of 12%.

Final answers: Average error rate = 9%; average accuracy = 91%; best fold = Fold 3; worst fold = Fold 4.

Problem 5: Application-Based Problem (10 Marks)

Question: A college develops a machine learning model to predict whether students will pass or fail an examination. The dataset contains 1,000 student records. The researchers use 10-fold cross-validation. The fold accuracies are 82%, 85%, 88%, 84%, 90%, 86%, 87%, 83%, 89%, and 86%.

1. How is the dataset divided? The 1,000 records are divided into 10 approximately equal folds. Records per fold = 1,000 / 10 = **100**.

2. Training and testing dataset sizes: Testing = 100 records; training = 1,000 – 100 = **900 records** per iteration.

3. Number of model training runs: The model is trained **10 times**.

4. Average accuracy:
(82 + 85 + 88 + 84 + 90 + 86 + 87 + 83 + 89 + 86) / 10 = 860 / 10 = **86%**.

5. Training accuracy is 95%, but average cross-validation accuracy is 86%. What does this indicate?
The 9-percentage-point gap may indicate overfitting. The model performs very well on training data but less well on held-out data. Possible solutions include regularization, reducing model complexity, collecting more representative data, feature selection, and appropriate hyperparameter tuning.

6. Two advantages of K-fold cross-validation:
• Better use of data: every record is used for testing exactly once and for training in the other K – 1 iterations, assuming standard K-fold partitioning.
• More reliable evaluation: performance is measured across multiple data splits, reducing dependence on a single train-test split.

Quick Revision: Important Formulas

Parameter	Formula
Records per fold	N / K
Training records per fold	N – N/K

Parameter	Formula
Average accuracy	Sum of fold accuracies / K
Average error rate	Sum of fold error rates / K
Average accuracy from error rate	100% – average error rate

Here, N is the total number of records and K is the number of folds.

Important note: These calculations assume equal-sized folds. For classification tasks, stratified K-fold cross-validation is often preferred because it preserves the approximate class proportions in each fold.